1.2 - Un-blinding the Operator: the PerfGD-Corrected Loop

REFLEX project - `math-theory`

STATUS: DONE (Priority 2, derivation/theorem). This document discharges the mathematical content of methodology target **1.2**. It builds on the closed forms of `01-analytic-stability-boundary` to derive the Performative Gradient Descent (PerfGD) correction in closed form, prove it converges to the *performative optimum* (PO) rather than the *stable point* (SP), show it remains stable for ε beyond the RRM boundary $\varepsilon^* = \gamma/\beta$, and quantify the echo-chamber gap. The remaining code task - wiring this gradient into a training loop as `equilibrium/perfgd_loop.py` - is called out in §8 and is out of scope for the math-theory deliverable.

One-line thesis. Blind RRM finds the spread that is its own best response given the induced flow - the stable point. It is blind to the fact that its own quoting moves that flow. PerfGD adds exactly the missing term $(\partial J/\partial T)(d\tau/dh)$, which 1.1 already gives in closed form, so no estimation is needed. The corrected loop optimises the true objective and is governed by the objective's own curvature γ_{PO}, not by the cobweb modulus $\varepsilon\beta/\gamma$ - which is why it converges where RRM oscillates and diverges.

1 Carry-over and new notation

From `01-analytic-stability-boundary` we reuse the expected one-step objective at frozen toxic level T,

$$J(h;T) = P\left[\,hAe^{-kh}\rho + hT - \psi T - w(h-2-\lambda_q\,\right], \qquad P = \texttt{pnl\_scale},$$

and the three closed-form constants

$$\gamma = -\frac{\partial^2 J}{\partial h^2} = P\left[\,2w + A\rho k e^{-kh}(2-kh) + \lambda_q\,\right] \qquad \text{(strong convexity)},$$

$$\beta = \left|\frac{\partial^2 J}{\partial h\,\partial T}\right| = P \qquad \text{(joint smoothness)},$$

$$\varepsilon = \left|\frac{d\tau}{dh}\right| = \rho\bar{g}\alpha f I c_t e^{-c_t h} \qquad \text{(distribution sensitivity)},$$

with $\tau(h) = \rho\bar{g}(I_b + \alpha f I e^{-c_t h}) = C_0 + C_1 e^{-c_t h}$, $C_0 = \rho\bar{g}I_b$, $C_1 = \rho\bar{g}\alpha f I$, and $\varepsilon(h) = c_t C_1 e^{-c_t h}$.

New symbol	Reading	Definition
$\Phi(h)$	performative objective (maximise)	$\Phi(h) = J(h; \tau(h)) = \mathbb{E}_{D(h)}[\mathrm{obj}]$
$\mathrm{PR}(h)$	performative risk (minimise)	$\mathrm{PR}(h) = -\Phi(h)$
h_{SP}	stable point (RRM fixed point)	zero of the blind gradient, §2
h_{PO}	performative optimum	maximiser of Φ, §2
$G(h)$	blind / decoupled gradient	$G(h) = \partial_1 J(h; \tau(h))$
$\Delta(h)$	performative correction	$\Delta(h) = \partial_T J \cdot (d\tau/dh)$
γ_{PO}	objective curvature at the PO	$\gamma_{\mathrm{PO}} = -\Phi''(h_{\mathrm{PO}})$, §4
ψ	per-unit adverse severity	$\psi = \sigma_s u\, a(u)/\bar{g}(u) > 0$

2 Two solution concepts

Stable point h_{SP} (what RRM finds). The dealer best-responds to the frozen distribution, then redeploys. A fixed point zeroes the blind gradient - the derivative of J in its first (decision) argument only, then evaluated self-consistently at $T = \tau(h)$:

$$G(h) := \partial_1 J(h; \tau(h)) = P\left[\, A\rho e^{-kh}(1 - kh) + \tau(h) - 2w(h - \,\right], \qquad G(h_{\mathrm{SP}}) = 0. \tag{1}$$

This is the FOC of 1.1; h_{SP} is the h^* of that document.

Performative optimum h_{PO} (what we want). The maximiser of $\Phi(h) = J(h; \tau(h))$, whose stationarity uses the total derivative:

$$\Phi'(h) = G(h) + \Delta(h), \qquad \Delta(h) := \partial_T J(h; \tau(h)) \cdot \frac{d\tau}{dh}, \qquad \Phi'(h_{\mathrm{PO}}) = 0. \tag{2}$$

The only difference between the two FOCs is Δ, and 1.1 already gives every piece of it.

3 The analytic performative gradient (no estimation)

(a) Marginal value of toxic flow. The toxic spread-capture term hT contributes $+h$; the adverse term $-\psi T$ contributes $-\psi$ (constant in h, but not in T, so it returns here):

$$\partial_T J = P\,(h - \psi).$$

A marginal unit of toxic flow earns half-spread h but costs ψ to adverse selection; net value $h - \psi$. Toxic flow is net profitable where $h > \psi$.

(b) Flow response $d\tau/dh = -\varepsilon(h)$ (from 1.1). Hence

$$\Delta(h) = \partial_T J \cdot \frac{d\tau}{dh} = P(h - \psi)\big(-\varepsilon(h)\big) = -P(h - \psi)\,\varepsilon(h), \tag{3}$$

and the full PerfGD ascent direction is

$$\Phi'(h) = G(h) + \Delta(h) = P\left[\, A\rho e^{-kh}(1 - kh) + \tau(h) - 2w(h - - (h - \psi)\varepsilon(h)\,\right]. \tag{4}$$

Every term is closed-form in the config and h. No finite-difference probing of $dD/d\phi$ is required - the contrast with Izzo et al. (2021), who estimate the distribution-response and inherit its bias and variance. Here 1.1 supplies it exactly.

4 The PerfGD update and its convergence

$$\text{PerfGD:} \qquad h_{k+1} = h_k + \eta\,\Phi'(h_k) \quad \text{(gradient ascent on the true objective).} \tag{5}$$

Contrast the two loops:

$$\begin{aligned}
\text{RRM / blind:} \quad & h_{k+1} = h_k + \eta\, G(h_k) & \to\ h_{\mathrm{SP}} \quad (\text{if } \varepsilon < \gamma/\beta),\\
\text{PerfGD:} \quad & h_{k+1} = h_k + \eta\,[G(h_k) + \Delta(h_k)] & \to\ h_{\mathrm{PO}} \quad (\text{under §4.3}).
\end{aligned}$$

The only added work is the scalar $\Delta(h_k) = -P(h_k - \psi)\varepsilon(h_k)$, so PerfGD costs the same per step as RRM.

4.1 The objective curvature γ_{PO}

Differentiate $\Phi(h) = J(h; \tau(h))$ twice, using $\tau' = -\varepsilon$, $\tau'' = c_t\varepsilon$:

$$\begin{aligned}
\text{uninformed: } & \tfrac{d^2}{dh^2}[hA\rho e^{-kh}] = -A\rho k e^{-kh}(2 - kh),\\
\text{toxic capture: } & \tfrac{d^2}{dh^2}[h\tau(h)] = 2\tau' + h\tau'' = \varepsilon(c_t h - 2),\\
\text{adverse: } & \tfrac{d^2}{dh^2}[-\psi\tau(h)] = -\psi c_t \varepsilon,\\
\text{quoting cost: } & \tfrac{d^2}{dh^2}[-w(h-2] = -2w.
\end{aligned}$$

Summing and multiplying by P,

$$\Phi''(h) = -\gamma + \beta\varepsilon\,(c_t h - 2 - c_t\psi),$$

so the objective curvature at the optimum is

$$\boxed{\gamma_{\mathrm{PO}} := -\Phi''(h_{\mathrm{PO}}) = \gamma + \beta\varepsilon\,(2 + c_t\psi - c_t h_{\mathrm{PO}}).} \tag{6}$$

4.2 Rate

If Φ is L-smooth ($|\Phi''| \le L$) and concave, gradient ascent with $\eta \le 1/L$ gives the standard convex rate on the value gap,

$$\Phi(h_{\mathrm{PO}}) - \Phi(h_k) \le \frac{(h_0 - h_{\mathrm{PO}})^2}{2\eta k} = O(1/k),$$

matching the stated $O(1/k)$. If additionally $\gamma_{\mathrm{PO}} > 0$ (§4.3), the rate upgrades to linear:

$$|h_k - h_{\mathrm{PO}}| \le (1 - \eta\gamma_{\mathrm{PO}})^k\,|h_0 - h_{\mathrm{PO}}|, \qquad \eta \le 1/L.$$

4.3 When is Φ strongly concave?

From (6), Φ is strongly concave at the optimum iff $\gamma + \beta\varepsilon(2 + c_t\psi - c_t h_{\mathrm{PO}}) > 0$. A clean sufficient condition is

$$c_t h_{\mathrm{PO}} < 2 + c_t\psi,$$

under which $\gamma_{\mathrm{PO}} > \gamma > 0$ for every ε (the correction only adds curvature).

5 Headline: PerfGD is stable beyond the RRM boundary

Theorem 1 (stability beyond ε^*)**.** *The two stability conditions are governed by different quantities:*

$$\begin{aligned} \text{RRM stable} &\iff m = \varepsilon\beta/\gamma < 1 \iff \varepsilon < \varepsilon^* := \gamma/\beta, \\ \text{PerfGD stable} &\iff \gamma_{\text{PO}} > 0 \text{ and } \eta < 2/L. \end{aligned}$$

Under the condition $c_t h_{\text{PO}} < 2 + c_t\psi$ of §4.3, $\gamma_{\text{PO}} = \gamma + (\text{positive}) \cdot \varepsilon > 0$ for all $\varepsilon \geq 0$, so PerfGD converges for arbitrarily large ε, including ε well beyond ε^* where RRM has already diverged.

Intuition: RRM diverges not because the objective lost its optimum but because the fixed-point iteration overshoots it (a negative-slope cobweb, 1.1 §3.3); PerfGD descends the objective directly and never iterates the cobweb, so the cobweb modulus is irrelevant to it.

6 The echo-chamber gap (stable vs. optimal)

Linearise the PO FOC about the SP. With $G(h_{\text{SP}}) = 0$ and $G'(h_{\text{SP}}) = \partial_1\partial_1 J + \partial_1\partial_T J \cdot \tau' = -\gamma - \beta\varepsilon = -(\gamma + \beta\varepsilon)$:

$$0 = G(h_{\text{PO}}) + \Delta(h_{\text{PO}}) \approx -(\gamma + \beta\varepsilon)(h_{\text{PO}} - h_{\text{SP}}) + \Delta(h_{\text{SP}}).$$

With $\Delta(h_{\text{SP}}) = -\beta\varepsilon(h_{\text{SP}} - \psi)$,

$$\boxed{h_{\text{SP}} - h_{\text{PO}} = \frac{\beta\varepsilon(h_{\text{SP}} - \psi)}{\gamma + \beta\varepsilon} = O(\varepsilon).} \tag{7}$$

When toxic flow is net profitable at the stable spread ($h_{\text{SP}} > \psi$), the stable point quotes strictly wider than the optimum: the dealer over-defends because, blind to $dD/d\phi$, it never credits itself for the flow its own tightening would summon. The value loss is quadratically smaller, since $\Phi'(h_{\text{PO}}) = 0$:

$$\boxed{\Phi(h_{\text{PO}}) - \Phi(h_{\text{SP}}) \approx \tfrac{1}{2}\gamma_{\text{PO}}(h_{\text{SP}} - h_{\text{PO}})^2 = \tfrac{1}{2}\gamma_{\text{PO}}\left[\tfrac{\beta\varepsilon(h_{\text{SP}} - \psi)}{\gamma + \beta\varepsilon}\right]^2 = O(\varepsilon^2).} \tag{8}$$

Reconciliation. The methodology README quotes the echo-chamber gap as $O(\varepsilon^2)$. That is the value gap (8) - the standard performative-prediction suboptimality result. The raw spread inflation (7) is $O(\varepsilon)$ to leading order. We report both and label which is which; conflating them is a common slip.

7 Corollaries

Corollary 1 (the correction is self-financing at $h = \psi$)**.** *$\Delta(h) = -\beta(h - \psi)\varepsilon(h)$ vanishes at $h = \psi$: when the quoted half-spread equals the adverse severity, toxic flow is value-neutral and PerfGD and RRM momentarily agree. The correction flips sign there - pulling toward tighter quotes when $h > \psi$ (chase profitable summoned flow) and wider when $h < \psi$ (suppress toxic summoned flow).*

Corollary 2 (gap scales linearly in adversariality)**.** *Since $\varepsilon \propto \alpha f$, the decision gap (7) obeys $h_{\text{SP}} - h_{\text{PO}} \approx (\alpha f)\,\rho\bar{g}Ic_t e^{-c_t h_{\text{SP}}}(h_{\text{SP}} - \psi)/\gamma$ to leading order: the echo chamber widens linearly as the market gets more adversarial.*

Corollary 3 (PerfGD vs. RRM curvature, side by side)**.**

$$\gamma_{\text{eff}}^{\text{RRM}} = \gamma - \beta\varepsilon, \qquad \gamma_{\text{PO}} = \gamma + \beta\varepsilon(2 + c_t\psi - c_t h_{\text{PO}}).$$

The ε-coefficients differ in sign and size: RRM loses curvature as feedback grows, PerfGD (in the operating regime) gains it. This is the formal statement of "un-blinding stabilises the loop".

8 Validation protocol and the remaining code task

Validation. With the same harness as 1.1 §7:

1. Compute h_{SP} (1.1 fixed point) and h_{PO} (root of Φ' in (4)), then the gaps (7), (8), across `sweep_feedback.yaml`.
2. Run RRM and PerfGD from a common start; confirm RRM diverges and PerfGD converges for ε just above $\varepsilon^* = \gamma/\beta$.
3. Confirm measured $h_{\text{SP}} - h_{\text{PO}}$ matches (7) ($O(\varepsilon)$) and the performative-risk gap matches (8) ($O(\varepsilon^2)$), with cross-seed IQR bands.

Remaining code task (out of scope here). Implement `equilibrium/perfgd_loop.py`: identical to the RRM loop but adding the scalar correction $\Delta(h_k) = -P(h_k - \psi)\varepsilon(h_k)$ to the policy gradient, with $\varepsilon(h)$ and ψ imported from a shared analytic module (the `analytic_boundary.py` of 1.1 §7). The math here fixes that module's formulas completely; no estimator is needed.

9 Honest caveats

- **$\gamma_{\text{PO}} > 0$ is conditional (§4.3).** In the defensive wide-spread regime $c_t h_{\text{PO}} > 2 + c_t\psi$ the performative second-order term turns destabilising and γ_{PO} can fall below γ; PerfGD then needs a smaller step or a trust region. The "stable beyond ε^*" claim carries this condition.
- **$\partial_T J = P(h - \psi)$ assumes the frozen-T objective of 1.1 (A3).** This is the first-order un-blinding (the leading $dD/d\phi$ term PerfGD targets); higher-order self-consistency (the full implicit $D = T(D, \phi)$ solve) is a further step.
- **Single bond, zero skew, reference-state $\psi, \bar{g}, \lambda_q$** - same scope limits as 1.1; the gap is a per-state quantity reported as a band over the reference-state distribution.
- **Linearised gap.** (7)/(8) are leading-order in ε; for large ε solve the two FOCs exactly (both are 1-D root-finds).

References

- J. Perdomo, T. Zrnic, C. Mendler-Dünner, M. Hardt. *Performative Prediction.* ICML 2020. - stable point vs. performative optimum; the $O(\varepsilon^2)$ suboptimality of the stable point.
- Z. Izzo, L. Ying, J. Zou. *How to Learn when Data Reacts to Your Model: Performative Gradient Descent.* ICML 2021. - the PerfGD algorithm, here run with the exact analytic $dD/d\phi$ from 1.1 instead of an estimated one.

- C. Mendler-Dünner, J. Perdomo, T. Zrnic, M. Hardt. *Stochastic Optimization for Performative Prediction.* NeurIPS 2020. - convergence rates for the corrected dynamics.

- Builds on `01-analytic-stability-boundary`; see `../references.bib` and `literature/literature-raghav/` for the full citation map.